The AI tools used for this project were Gemini, ChatGPT and NotebookLM. These were used to help with this project. These were chosen for a number of reasons, NotebookLM was especially used because it is very helpful to include research papers, provide summaries, generate videos and visualisations. Gemini and ChatGPT were used mainly to explain content and principles which were not understood. Multiple Gen AI tools were used to refine and improve the work. Using more than 1 tool allows to cross check provided outputs to reduce bias, this can also reduce the likelihood of hallucinations or factual errors. These tools were also used to help using new technologies.

The use of generative AI tools in this project required careful attention to ethical considerations, particularly in relation to data bias, accuracy, privacy, and academic integrity. While these tools can significantly enhance productivity and idea generation, they also introduce risks that must be actively managed. Generative AI models are trained on large datasets that may contain inherent social, cultural, or contextual biases. As a result, outputs may unintentionally reflect or reinforce these biases. To mitigate this, responses generated by AI were critically evaluated, and where necessary, adjusted to ensure neutrality, inclusivity, and alignment with credible academic perspectives. A significant ethical risk when using these tools is that these models sometimes hallucinate and generate confident but incorrect information. As mentioned above outputs were checked and validated using other sources and other Gen AI tools. Research papers were read and annotated to confirm summaries provided by both NotebookLM and the Gen AI tools. To verify accuracy, important claims, definitions, and explanations were cross-checked against reliable academic sources such as peer-reviewed journals, textbooks, and reputable websites and were not just accepted at face value.

Privacy was also considered during the use of AI tools. No personal, sensitive, or confidential information was entered into any generative AI system. All prompts were designed to be generic and task-focused, ensuring compliance with data protection principles and reducing the risk of unintended data exposure. My approach to academic conduct was to use these AI tools as aids for brainstorming, explaining, structuring, and refining content, rather than as a substitute for doing the work. Where AI contributed to the development of ideas or text, this use was transparently acknowledged.

When working on the study guide, Gemini was used to help draft a structured format of headers to optimise the flow to make understanding the content more clear. Using this approach it was decided that the first section would be explaining Softmax and Cross-entropy loss. To understand clearly how backpropagation and gradient flow works in CNNs, one needs to understand how the loss is calculated before being moved backward throughout the network. Using these tools the sections were drafted as to start from the loss calculations and move deeper into the backpropagation. Gemini was also used to help understand the softmax and cross-entropy formulas, since personally I’m not that good at Maths. Prompts were used to help understand all the terms in the equations, why the exponents were used and why use Euler’s constant.

When working on the second section of the study guide, which is the backpropagation section, both Gemini and ChatGPT were used. They were used to understand how backpropagation exactly works by combining what I already knew to the new concepts I needed to understand. I find that visual information is very helpful to understand concepts which are new to me, so I prompted to generate some images to help understand gradient flow. Unfortunately all the images that were being generated were too complicated and included a lot of text. However when further prompting to refine the image to make it more like I wanted it to be, it was more difficult and time consuming than I had intended it to be so instead images on the web were sourced and used.

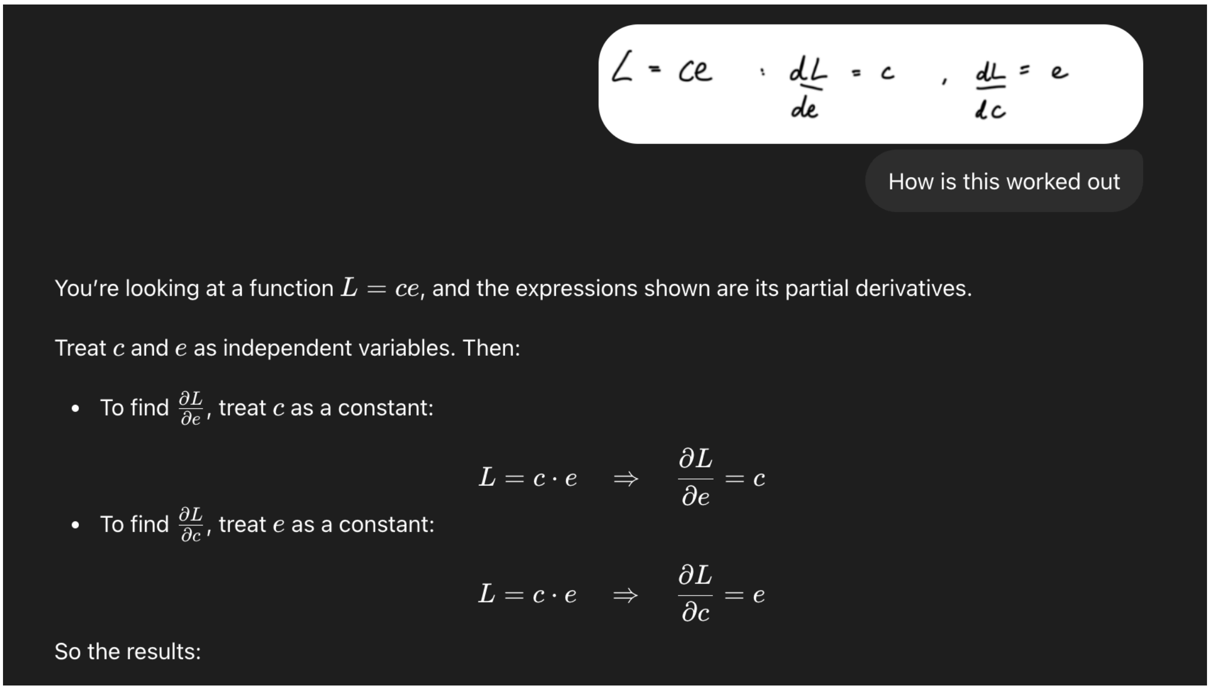
What I found very effective when using Gen AI tools was to understand the mathematical foundation of backpropagation and gradient flow. It has been a few years since I had learned and last used differentiation, so I had almost completely forgotten how it worked. Through research using academic sources and reputable websites it was understood that backpropagation is almost entirely differentiation since to optimise the loss, the weights are adjusted and the derivatives tell us how much a small change in weights can affect the loss calculated at the end. Due to this these tools were used to refresh how differentiation works and advance my understanding of how the chain rule works and how it is applied in backpropagation. Throughout this section, gen AI tools were mainly used to understand and help explain concepts which could not be easily understood from online sources and academic work.

When finding initial research papers to use in this project, Gen AI tools were used to generate ideas and recommendations on papers to search and read. As mentioned previously, these papers were collated in NotebookLM to keep track of and summaries. Other features which NotebookLM provides were used such as Infographic and video overview to further visualise and understand the papers.

NotebookLM along with Gemini was also used to help generate questions for the adversarial quiz. These tools were used to generate questions other than coming up with them myself for a couple of reasons. Mainly, using these tools allows me to revise and confirm the understanding of the topic so far. I used both of these tools to generate questions and answer them, this allowed me to revise the topic and confirm that I understood what I had covered. Since the questions were also going to be answered by my classmates, I had to update and add more information to the study guide to make sure that all the questions that were generated had answers and explanations which were covered in the study guide. This iterative process allowed the study guide to be refined as much as possible.

Gemini and ChatGPT were used as well to generate some code for the Interacative simulator. Implementing the simulator was the first time I used Streamlit so along with the documentations, these tools assisted in developing the application. The documentation was helpful when to set up the application, installing the packages and running the initial web based application locally. The Gen AI tools were prompted to generate some code based on a basic idea I had on how the simulator should be. From that response some code was copied and modified to better suit the application and user experience. Throught this work, Gemini was also used to refne wording and correcting sentence and paragraph wording.

Some prompts which meaningfully contributed to this are the following. When revising differentiation and I needed to remember how partial derivatives are worked, I found an example online which was covering some examples but did not provide a full worked example with explanations. The following prompt was helpful to explain how partial derivatives worked, and this allowed to keep building the mathematical foundation to continue work on the chosen topic.



$L = ce$: $\frac{dL}{de} = c$, $\frac{dL}{dc} = e$

How is this worked out

You're looking at a function $L = ce$, and the expressions shown are its partial derivatives.

Treat c and e as independent variables. Then:

- To find $\frac{\partial L}{\partial e}$, treat c as a constant:

$$L = c \cdot e \quad \Rightarrow \quad \frac{\partial L}{\partial e} = c$$

- To find $\frac{\partial L}{\partial c}$, treat e as a constant:

$$L = c \cdot e \quad \Rightarrow \quad \frac{\partial L}{\partial c} = e$$

So the results:

The following example is a prompt which was also very helpful to this work, when working on this project I found that I would constantly be studying and researching concepts and techniques which were new to me. These Gen AI tools are very good at explaining complex concepts, so throughout the process I used these tools to check and confirm what I had done. This self checking and validation allowed me to work confidently and correct where I did not understand.

and i assume only the final layer has the dL/dA since the loss is caluclated at the end

You are exactly right—the process has to start somewhere!

The very first "spark" that kicks off the entire backpropagation process happens at the **Loss Function** (L). Only the final layer of the network (usually a Softmax or Linear layer) calculates $\frac{\partial L}{\partial A}$ directly from the error.

Think of the Loss Function as the "Source" of the gradient. Once that first derivative is calculated, it flows backward like water through a pipe.

Some prompts which I also used which helped throughout this work was using these tools as a starting point. Instead of using a simple Google search and start going through the links, these tools were used to promt for topics or searching point which can be further explored. This method was also used to help find research papers which are related to the research topics. The following prompt shows and example of these prompts. I prompted what I wanted to argument and from the list of research papers which Gemini recommended I reviewed and validated the papers to use in the Key papers section of the study guide.

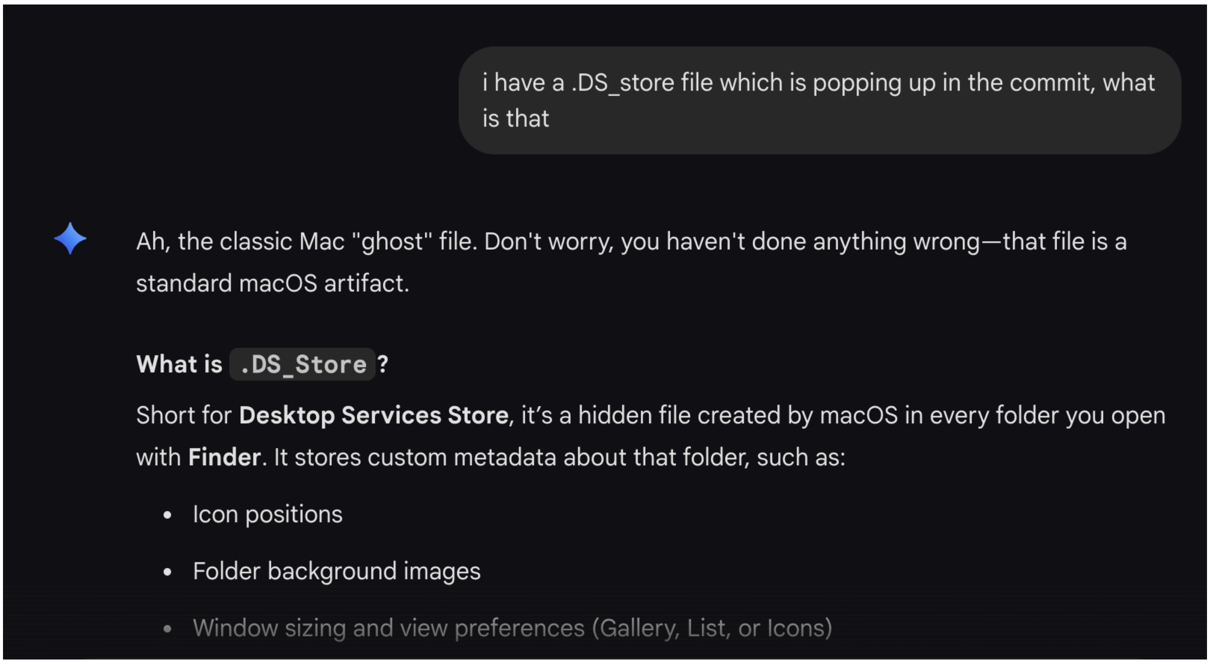
I want to demonstrate the growth of this topic, can i use the forward forward architecture to argue the future

Using Geoffrey Hinton's **Forward-Forward (FF)** algorithm (2022) is an excellent way to argue for the future of the topic. If the 1989 and 1998 papers represent the "Age of Backpropagation," the Forward-Forward algorithm represents a potential shift toward **biologically plausible** and **hardware-efficient** AI. +1

How to use it to argue the "Future":

- **Replacing the Backward Pass:** You can argue that the future may move away from the "backward locking" of the chain rule. While backpropagation requires a full forward pass to be completed before the backward pass can even begin, FF uses two forward passes (one "positive" with real data and one "negative" with corrupted data) to update weights locally. This allows for **asynchronous, parallel training** of layers. +1

Gemini was also used to help debug issues when setting up environment and issues running the interactive simulator. I recently switched from a Windows device to Mac, since this was the first time setting up github and running an application from the command line, I had some issues and concerns with the hidden files and active directories.

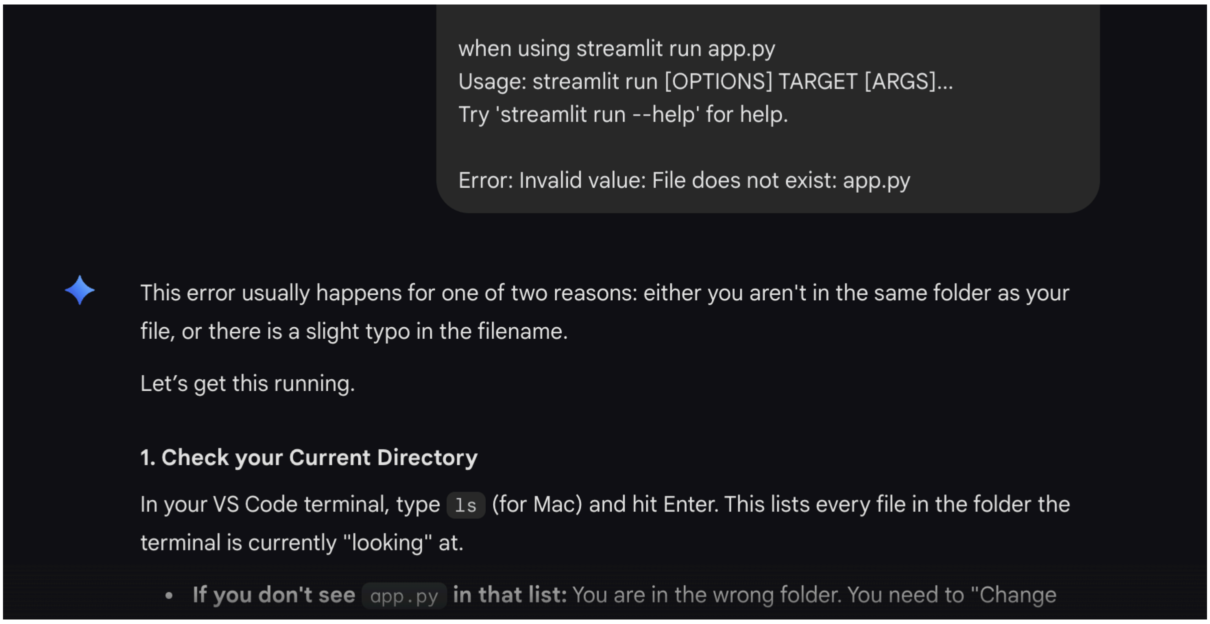

i have a .DS_store file which is popping up in the commit, what is that

Ah, the classic Mac "ghost" file. Don't worry, you haven't done anything wrong—that file is a standard macOS artifact.

What is `.DS_Store`?

Short for **Desktop Services Store**, it's a hidden file created by macOS in every folder you open with **Finder**. It stores custom metadata about that folder, such as:

- Icon positions
- Folder background images
- Window sizing and view preferences (Gallery, List, or Icons)


when using streamlit run app.py
Usage: streamlit run [OPTIONS] TARGET [ARGS]...
Try 'streamlit run --help' for help.

Error: Invalid value: File does not exist: app.py

This error usually happens for one of two reasons: either you aren't in the same folder as your file, or there is a slight typo in the filename.

Let's get this running.

1. Check your Current Directory

In your VS Code terminal, type `ls` (for Mac) and hit Enter. This lists every file in the folder the terminal is currently "looking" at.

- **If you don't see `app.py` in that list:** You are in the wrong folder. You need to "Change

As mentioned, Gemini was also used to help gerate some code and debug the interactive simulator. The initial prompt was what I wanted to achieve and from the initial response on how to do certain things such as the tabs and latex statements, the code was modified and moulded to be as I imaged the interactive simulator.

i want to make a computional graph, where 3 inputs are passes and the visualisation of the forward pass and then the backward pass

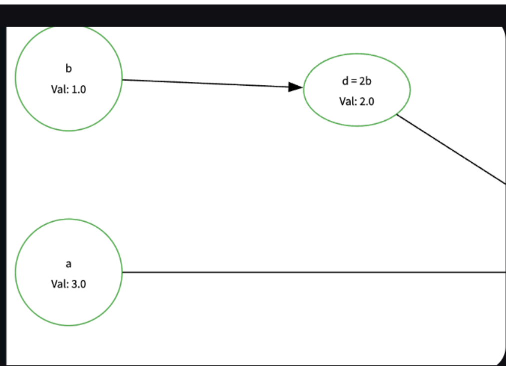
To build a computational graph in Streamlit that visualizes both the **Forward Pass** (calculating results) and the **Backward Pass** (calculating gradients) using the chain rule, the most efficient approach is to use `graphviz`.

Here is a plan to build this specific feature, focusing on a function like $f(x, w, b) = (x \cdot w) + b$.

1. The Logic Engine

You need to manually define the forward and backward passes so you can display the

Some prompts which helped to solve some issues with the graphs were the inputs were not being in alphabetical order and the graph was being aligned to the left margin and I wanted it to be center aligned.



a and b are flipped,

Analysis

Ah, the classic Graphviz "logic" where it tries to optimize space by rearranging nodes! Since a and b are independent inputs, it's just guessing where to put them.

To fix this and force the order to be a (top), b (middle), and c (bottom) exactly like your sketch how can i center align the graph

Streamlit's `st.graphviz_chart` is wrapped in a container that defaults to the left side of the column. To get that perfectly centered, professional "simulator" look, you have two main options:

1. The "Column Sandwich" (Recommended)

The cleanest way to center any element in Streamlit without custom CSS is to create three columns and place your graph in the middle one. You can adjust the ratios to make the center column wider or narrower.